

2-Wheel FPV Balancing Robot - V1 PoC BOM

Simplified build reference for the Rainbow Six Siege-inspired DJI O3 scout bot

Version: V1 PoC | Scope: RC FPV + balancing research | Jetson/ROS2 removed from V1

1. V1 Goal and Scope

V1 is a practical proof-of-concept for a compact 2-wheel FPV robot inspired by the Rainbow Six Siege drone. The build keeps the parts already available - 4S LiPo, IBT-2/BTS7960 drivers, Waveshare encoder motors, Teensy, ELRS, and DJI O3 - even though this makes the chassis bulkier. The goal is to validate drivetrain, FPV packaging, RC control, encoder readout, IMU angle estimation, and later balance PID before designing a smaller V2.

- Drive style: 2-wheel differential drive.
- FPV payload: DJI O3 Air Unit and camera.
- Control: ELRS receiver into Teensy 4.1.
- Motor control: one IBT-2/BTS7960 H-bridge per motor.
- Balancing: research target, but V1 may use a belly skid or hidden roller during early testing.
- Excluded from V1: Jetson Nano, onboard ROS2 compute, advanced autonomy, extra navigation sensors, and compact final shell optimization.

2. Simplified V1 BOM

Category	Component	V1 Decision	Notes
Drive	2x Waveshare DCGM-3865-12V-EN-240RPM	Keep	12V metal geared DC motors with integrated quadrature encoders.
Wheels	2x 90-100 mm grippy wheels	Add/confirm	Use normal rubber/TPU tires. Avoid mecanum for V1 balancing.
Motor Drivers	2x IBT-2 / BTS7960 H-bridge modules	Keep	One driver per motor. Bulky but acceptable for PoC.
Battery	4S LiPo, 14.8V nominal, 16.8V full, ~2000 mAh	Keep	Use conservative PWM limits because motors are rated for 12V.
Power Harness	XT60, inline fuse, rocker switch, WAGO-style distribution	Keep	Use 10-15A automotive blade fuse; start with 10A if practical.
Logic Buck	LM2596 DC-DC buck converter	Keep for PoC	4S to 5V for Teensy, ELRS, IMU, and encoder logic. Not for motors or O3.
O3 Power	Dedicated BEC/regulator for DJI O3	Keep/add	Prefer separate clean power path for O3 rather than sharing noisy motor wiring.
Controller	Teensy 4.1	Keep	Main real-time controller: PWM, encoders, IMU, ELRS CRSF, failsafe, later balance PID.
IMU	GY-91	Keep	Use gyro + accelerometer first. Magnetometer/barometer not needed in V1.
Radio	2.4 GHz ELRS receiver	Keep	UART CRSF at 420000 baud. Place upper/rear, away from motor power wiring.
FPV	DJI O3 Air Unit + camera + antennas	Keep	Camera front; air unit upper/rear with airflow; antennas rear/top.
Stability	Belly skid or hidden roller	Add	Useful during pre-balance RC testing and as crash

			protection.
Cooling	Side vents + rear/top exhaust + optional 25-30 mm fan mount	Add/confirm	Important because O3 and IBT-2 drivers can heat up.
Compute	Jetson Nano / ROS2 onboard compute	Remove	Not part of V1; reserve space only if easy.

3. Electrical Architecture

Recommended V1 power/control split:

4S LiPo

- > fuse + rocker switch
- > IBT-2 #1 -> left Waveshare motor
- > IBT-2 #2 -> right Waveshare motor
- > LM2596 5V buck -> Teensy + ELRS + GY-91 + encoder logic
- > dedicated BEC/regulator -> DJI O3 Air Unit

- Arduino Uno can be used only as a temporary motor/driver tester while waiting for the Teensy.
- All logic grounds must share a common reference with the motor driver ground.
- Keep high-current motor wiring short and physically separate from signal wiring where practical.
- Add 470-1000 uF electrolytic capacitance across each driver power input and 100 nF ceramic capacitors near motor terminals if available.
- With 4S on 12V motors, begin with low PWM limits. Initial software cap: 30-50 percent for bench tests, then increase carefully after checking heat and current.

4. Motor Reference

Parameter	Reference value
Rated voltage	12V
No-load output speed	~240 RPM
Loaded output speed	~125 RPM
Rated torque	~4 kg*cm = ~0.39 N*m
Stall torque	>=8.5 kg*cm = >=0.83 N*m
Encoder	Integrated quadrature encoder
Practical wheel diameter	80-120 mm; target 90-100 mm for V1

With a 100 mm wheel, rated torque gives roughly 7.8 N push force per wheel, or about 15.6 N total for two wheels before losses. Use this as a rough sizing reference, not a guaranteed traction value.

5. Mechanical Layout

TOP FRONT:

DJI O3 camera in protected nose mount

UPPER REAR:

DJI O3 Air Unit, O3 antennas, ELRS antenna, exhaust/vent area

CENTER:

Teensy 4.1 and GY-91 IMU on a rigid internal shelf

AROUND / ABOVE AXLE:

4S battery, central and ideally height-adjustable for balancing experiments

SIDES:

Waveshare motors and wheels

SIDE / REAR:

IBT-2 drivers with airflow and service access

BOTTOM:

Belly skid or sacrificial protection plate

- For balancing research, the center of mass should be above the wheel axle, but not extremely high.
- Make the battery central and height-adjustable if possible: low position for passive FPV tests, raised/axle-level position for balancing experiments.
- Use normal grippy wheels, not mecanum. Mecanum rollers make balancing response less predictable.
- The O3 camera should be front-facing, protected, and not blocked by the body or wheels.
- The O3 air unit should not be buried in a sealed shell; reserve airflow and service access from the beginning.

6. Fusion 360 Component Structure

Robot_V1_PoC

00_Reference_Skeleton

- ground plane
- wheel axle axis
- center symmetry plane
- target center-of-mass zone

10_Lower_Tub

- motor mounts
- battery rails
- belly skid
- power cable channels

20_Upper_Shell

- O3 bay
- service hatch
- cooling vents
- antenna exits

30_Drivetrain

- left motor module
- right motor module
- wheels and hubs

40_Electronics

- Teensy shelf
- IMU mount
- ELRS mount
- buck converters

50_Motor_Drivers

- left IBT-2 module
- right IBT-2 module
- airflow clearances

60_FPV

- O3 camera nose
- O3 Air Unit mount
- antenna guards

7. V1 Development Order

Phase	Task	Purpose
1	Power validation	Verify 4S, fuse, switch, 5V buck, and O3 regulator before connecting sensitive electronics.
2	Motor PWM test	Use Arduino Uno or Teensy to drive one IBT-2 and one motor at low PWM.
3	Encoder readout	Confirm encoder A/B wiring, count direction, and RPM calculation.
4	Wheel velocity PID	Control target RPM before attempting balance.
5	IMU angle estimation	Use GY-91 gyro + accelerometer with complementary filter.

6	Balance PID	Start on a stand, then tethered floor tests with low PWM limit and cutoff angle.
7	ELRS integration	Map throttle/steering/arm switch to desired speed, turn, and failsafe.
8	DJI O3 integration	Install O3 after chassis can drive safely; verify cooling and antenna placement.

8. Early Test Limits and Safety

- Remove wheels for the first motor tests.
- Start with PWM values around 40-80 out of 255 when using Arduino analogWrite on 4S.
- Do not test 100 percent PWM on 4S until motor temperature and current are understood.
- Use an inline fuse and a real power switch during bench tests.
- Add a software arming state before motors can move.
- For balance PID, add a cutoff if body angle exceeds roughly 25-35 degrees.
- Test O3 installation only after the drivetrain is mechanically safe and protected.

9. Removed or Deferred from V1

- Jetson Nano and onboard ROS2 compute.
- Advanced autonomy, mapping, AI vision, and navigation stack.
- Mecanum wheels.
- Sonar/ToF side sensors unless already available and easy to mount.
- Final compact shell optimization.
- Full EMI/vibration refinement - acceptable to postpone for balancing research, but leave space for later improvements.